

1 MYB regulator of 'colorless' flavonols underlies the evolution of red flowers in *Lochroma*
2 (Solanaceae)

3
4 Lucas C. Wheeler^{1,†}, Maximilian Larter², Stacey D. Smith^{1,†}

5 ¹Department of Ecology and Evolutionary Biology, University of Colorado Boulder, Boulder, CO
6 80309, USA

7 ²National Research Institute for Agriculture, Food and Environment (INRAE), 33076 Bordeaux,
8 France

9 [†]authors for correspondence: lwheeler9@gmail.com, stacey.d.smith@colorado.edu

10
11 **Running head:** Novel R2R3-MYB regulator of red flowers

Abstract

Anthocyanins, the pigments that give rise to blue, purple, red and pink colors in many flowers and fruits, are produced by the deeply conserved flavonoid biosynthesis pathway. The regulation of this pathway is thus fundamental for species differences in color across flowering plants, and a growing body of evidence implicates MYB transcription factors as key players activating or suppressing the production of different pigments. Nevertheless, the potential role of MYBs genes in determining the type of pigment produced (as opposed to the overall amount) is unknown. Here we demonstrate that a lineage of R2R3 MYBs that is closely related to well-known flavonol regulators (MYB12 members in subgroup 7) is the primary determinant of the shift from blue to red flowers in the genus *Lochroma*. Similar to its ortholog in *Capsicum*, this *Lochroma* MYB12-like gene controls the expression of flavonoid-3'-hydroxylase, the pathway branch point between red and blue pigments, and when down-regulated, results in redirection of flux toward red pigments. These results underscore the importance of transcription factor evolution in generating phenotypic novelty as well as the competitive nature of interactions among flavonoid pathway branches. In addition, our study demonstrates the effectiveness of RNAseq of segregating populations, in combination with other lines of evidence, for identifying novel functional variation.

Keywords: Transcriptomics, flavonoid biosynthesis, pigmentation, flower color, pelargonidin, gene regulation

Introduction

Phenotypic differences between species are often controlled by differences in the timing and patterns of gene expression (Kimura *et al.* 2008; Des Marais and Rausher 2010; Byers *et al.* 2014). These differences in gene expression can arise through a variety of mechanisms, including changes in the *cis*-regulatory regions controlling expression (i.e., promoters, enhancers), changes in the expression or function of transcription factors, or post-transcriptional regulation (e.g., gene silencing). Many authors have argued that the *cis*-regulatory mutations will be favored during evolutionary transitions due to their modular architecture, allowing for altered expression in one context without pleiotropic effects in other contexts (Prud'homme *et al.* 2006; Wray 2007). However, functional changes in transcription factors can have similarly narrow consequences, depending on their specificity in terms of target genes and spatio-temporal patterns of expression (Lynch and Wagner 2008; Panchy *et al.* 2016; Auge *et al.* 2019).

Plant MYB transcription factors comprise a prime example of a large and diverse gene family with highly specialized functions. Whereas animal and fungal genomes house at most a few dozen MYB genes, plant genomes contain hundreds of MYBs, even in diploid species (Shiu *et al.* 2005; Feller *et al.* 2011; Gates *et al.* 2016). This expansion of MYB copies in plants is coupled with a diversification of functional roles, from defense, to coloration, to morphology (Ramsay and Glover 2005; Wu *et al.* 2022). Closely related MYBs often share similar regulatory functions, e.g., as activators or repressors of particular sets of target genes, but vary in their expression patterns, resulting in similar phenotypic effects albeit in different tissues or developmental stages (e.g., Millar and Gubler 2005; Stracke *et al.* 2007). Nevertheless, with the multitude of MYBs in every plant genome, new functional roles and patterns of diversification are continuing to be discovered (Sagawa *et al.* 2016; Gates *et al.* 2018; Mu *et al.* 2024).

Among the subgroups of plant MYB transcription factors, those regulating floral coloration through the production of flavonoid pigments are among the best studied. The MYB activators of flavonoid synthesis fall into several subgroups of R2R3 MYBs, including the subgroup 7 (SG7) genes that regulate the 'early' genes of the pathway (e.g., *CHS*, *F3H*) and the branches leading to flavonol production (*FLS*), the subgroup 6 (SG6) genes that regulate the 'late' steps of the pathway (e.g., *DFR*, *ANS*) leading to anthocyanin pigments, and the subgroup 5 (SG5) genes that control proanthocyanidin production (Gonzalez *et al.* 2009; Dubos *et al.* 2010; Albert *et al.* 2014) (Fig. 1). Anthocyanins give rise to the red, purple and blue floral hues, while flavonols can modify these colors as co-pigments and provide UV-absorbing patterns,

such as nectar guides and bullseyes (Sheehan *et al.* 2016; Todesco *et al.* 2022). Thus, both types of compounds (anthocyanins and flavonols) are important contributors to floral coloration and are often jointly produced in developing petals.

While this general regulatory architecture is well-conserved across flowering plants (Mol *et al.* 1998; Schwinn *et al.* 2014), the factors determining the type of flavonol or anthocyanin produced appear more variable across species, and perhaps for that reason, are not as well understood. Both flavonols and anthocyanins are produced at three hydroxylation levels (mono-, di-, and tri-) that have different spectral properties, and their relative expression depends on the expression of the so-called branching enzymes, F3'H and F3'5'H (Fig. 1). For example, when both enzymes are highly expressed, flowers will produce the tri-hydroxylated flavonoids, such as the blue delphinidin pigments, whereas when these enzymes are not present, flowers will produce the red pelargonidin pigments (Wessinger and Rausher 2012; Fig. 1). The F3'H enzyme, which is responsible for conversion of DHK (the precursor of the flavonol kaempferol and the red pigment pelargonidin) into DHQ (the precursor of the flavonol quercetin and the purple pigment cyanidin), appears to be regulated by subgroup 7 MYBs in *Capsicum* (Wu *et al.* 2023) and subgroup 6 MYBs in petunia and *Antirrhinum* (Schwinn *et al.* 2006; Albert *et al.* 2011). The other branching enzyme, F3'5'H, has been lost in many flowering plant lineages (e.g., morning glories, roses, penstemons) (Rausher 2006; Wessinger and Rausher 2012), but in those species which have an intact copy and have been well-studied in terms of anthocyanin regulation, its expression is typically co-regulated with the anthocyanin-specific steps of the pathway by the subgroup 6 MYBs (Albert *et al.* 2011).

Substrate preference of the multifunctional enzymes also plays a key role in the type of flavonoids produced. While several of the pathway enzymes are able to accept multiple precursors, they often exhibit higher activity for one or a subset of substrates. For example, the petunia DFR enzyme is able to act on all three dihydroflavonol precursors (DHK, DHQ, and DHM), it preferentially acts on the latter two to produce cyanidin and delphinidin-derived anthocyanins (Johnson *et al.* 1999). The FLS enzyme, which competes with DFR for dihydroflavonols, also acts as a substrate specialist in most taxa with its preference often overlapping that of DFR (Choudhary and Pucker 2024). Nevertheless, the available substrates for both enzymes depend on the activity of F3'H and F3'5'H, which are required for DHQ and DHM production. Moreover, the overlap in function between F3'H and F3'5'H depends on the species; in some taxa, such as the tea plant and some Asteraceae, F3'5'H can carry out both 3' and 5' hydroxylation (Seitz *et al.* 2007; Wang *et al.* 2014), while in others, such as petunia and *Lochroma*, this enzyme primarily acts at the 5' position to make DHM from DHQ (de Vetten *et al.*

1999; Smith and Rausher 2011; Fig. 1). Thus, the flux through the pathway and the resulting pigmentation depends on the complex interplay of patterns of expression and functionality of these enzymes (Wheeler et al. 2021; 2023).

In the present study, we investigate the regulatory control of *F3'H* expression in *lochroma* (nightshade family, Solanaceae), one of several genera in which red pelargonidin-producing flowers have evolved from blue delphinidin-producing ancestors. Previous work demonstrated that this flower color transition in the red-flowered lineage involved three genetic changes, including the floral down-regulation of *F3'H*, the loss of the *F3'5'H* gene and a shift to specialization of DFR on DHK (Smith and Rausher 2011; Smith et al. 2013). Among these changes, the loss of *F3'H* expression has the largest effect on floral pigmentation because it largely eliminates flux away from DHK, allowing anthocyanin production to be entirely redirected towards pelargonidin (Smith and Rausher 2011; Fig. 1, note that *F3'5'H* is specialized for DHQ in Solanaceae). Moreover, this shift in *F3'H* expression is due to a *trans*-regulatory mutation, as the genotype at the *F3'H* locus itself does not predict flower color in segregating populations (Smith and Rausher 2011). This unknown regulator of *F3'H*, which segregates as a single gene, was termed the '*T*-locus' (Smith and Rausher 2011).

Here we use a suite of genomic, transcriptomic, and biochemical approaches to identify candidates for the *T*-locus responsible for the shift toward pelargonidin production and in turn, the evolution of red flowers in *lochroma*. Using biochemical and expression data, we first sorted individuals from a backcross population by pigment phenotype and corresponding difference in *F3'H* expression. Next, we searched the floral transcriptomes of these two pools of individuals for genes that match the predicted allelic pattern (e.g., homozygous for the red-flowered parent allele in the pink/red-flowered pool) and show the predicted association with *F3'H* expression. Our analyses point to a single R2R3 MYB transcription factor that is related to the MYB12 members of Solanaceae subgroup 7 MYBs but falls in a deeply diverged clade, only functionally characterized in chili peppers. As we discuss, these results suggest that the subgroup 7 MYBs may be much more diverse than previously known and play an underappreciated role in flower color evolution through their effects on flavonol production.

Materials and Methods

Source populations and phenotyping

Individuals of the blue-flowered *I. cyaneum* were crossed with the red-flowered *I. gesnerioides* to create segregating populations to dissect the genetic basis of their flower color differences (Smith and Rausher 2011). The blue-flowered state is ancestral in *Lochroma* and corresponds to the production of delphinidin-derived anthocyanins, while the red-flowered derived state involves the production of pelargonidin-derived anthocyanins (Fig. 1; Smith and Rausher 2011). The *I. cyaneum* parent was grown from seed from a cultivated accession from the Missouri Botanical Gardens, originally collected by W. G. D'Arcy, and the *I. gesnerioides* parent was grown from the Solanaceae Germplasm collection in the Botanical Garden of Nijmegen (accession number 944750129). Herbarium vouchers for these accessions are Smith 265 and 266 (WIS), respectively. A single F1 was backcrossed to the *I. gesnerioides* parent, and progeny from the resulting backcross population were genotyped at *F3'5'h* and *Dfr* (Smith and Rausher 2011; Table 1). Anthocyanin production was previously characterized using HPLC and revealed three pigment phenotypes (purple-flowered individuals producing primarily cyanidin, pink-flowered individuals producing mostly pelargonidin, and red-flowered individuals producing almost entirely pelargonidin) (Smith and Rausher 2011). The purple-flowered individuals share high *F3'H* expression and are inferred to carry a dominant 'blue' allele at a segregating *trans*-acting factor (the '*T*-locus', Smith and Rausher 2011) (Table 1).

Biochemical phenotyping and RNA-seq of backcross individuals

We performed RNA-Seq on corolla tissue from 24 backcross individuals segregating for the putative *T*-locus. We sampled 12 individuals with each inferred *T*-locus genotype: *Tt* corresponding to one dominant 'blue' allele and high *F3'H* expression or *tt* corresponding to two recessive red alleles and low *F3'H* expression (Table 1). We divided these 12 among the possible genotypes at the other two loci that affect flower color in this cross (*DFR* and *F3'5'H*). *DFR* shows functional specialization, with the red allele specialized for activity on DHK (Smith *et al.* 2013), while *F3'5'H* is absent from the red parent genome (Smith and Rausher 2011). With four possible combinations at these other two loci (*Dd/F-*, *Ddl-*, *dd/F-*, *ddl-*), we sampled three biological replicates of each within the groups of 12 (Table 1). We included all possible genotypic combinations at the three loci influencing flower color in order to isolate the *T*-locus while balancing across the effects of these other loci. For RNA extraction, we flash-froze corolla tissue from buds of roughly 1.25cm in length, which is equivalent to petunia bud Stage 5 (Pollak *et al.* 1993). This developmental stage shows expression of both early and late pathway genes in the anthocyanin pathway (Larter *et al.* 2018). Total RNA was extracted with the Spectrum

Total RNA extraction kit (Sigma, St Louis, MO). Library preparation and 150-base-pair paired-end mRNA sequencing was carried out by Novogene (Sacramento, CA).

Identifying SNPs associated with flower color and *F3'H* expression

We used the reference genome assembly for *Lochroma cyaneum* (Powell *et al.* 2022) to call SNP variants and filter the RNASeq dataset for candidate genes for the *T*-locus. RNAseq reads were aligned with STAR v. 2.75b (Dobin *et al.* 2013), and the resulting BAM files were used as input for the mpileup tool in bcftools v. 1.17 (Danecek *et al.* 2021). We used the tools mpileup and call in bcftools with default settings to call allelic variants in expressed genes. We filtered variants by base call quality, only retaining variants with quality score greater than or equal to 20. We used the resulting VCF file for subsequent analyses of associations with the color phenotype.

We first split the filtered VCF files into two subsets, one for all samples with purple cyanidin-producing flowers (inferred *Tt* genotype at *T*-locus) and one for those with pink or red mostly pelargonidin-producing flowers (inferred *tt* genotype at *T*-locus) (Table 1, Fig. 1a). In order to identify SNPs that differ between these two pools, we used *pyvcf* (Casbon 2012) to filter the variants to include only those that are present in all “*Tt*” individuals and not present in any “*tt*” individuals. This strict criterion resulted in a set of SNPs that perfectly co-segregate with the high or low *F3h* expression (see Results). Most of the SNPs are located on chromosome 5, but some mapped to smaller scaffolds that were not incorporated into the reference assembly (Supplementary Fig. 1). We then used *promer* from Mummer4 (Marçais *et al.* 2018) and *D-genies* (Cabanettes and Klopp 2018) to align these scaffolds back to the *L. cyaneum* and tomato reference genomes.

In addition to this filtering approach, we performed a case-control GWAS with the variant calls in GEMMA 0.98.5 (Zhou and Stephens 2012). We set phenotypes to 0 (purple-flowered *Tt* plants) or 1 (pink/red-flowered *tt* plants) and fit a univariate linear mixed model with the full set of variants. We then plotted the location of all analyzed variants on the assembled *L. cyaneum* chromosomes and scanned variants with significant phenotypic associations using $P < 5 \times 10^{-8}$ as the genome-wide cutoff (Xu *et al.* 2014).

Co-expression of candidate genes with *F3'H*

We predicted that if the *T*-locus is a transcriptional regulator, its expression will likely track that of *F3'H* in the segregating backcross. Thus, we used expression data from the 24 transcriptomes to quantify levels of expression and test for correlations between *F3'H* and loci carrying associated SNPs (previous section). We first created a *de novo* transcriptome for the blue-flowered parent (*I. cyaneum*) to ensure that we captured all expressed genes. For this assembly, we used single-end Illumina RNA-seq data from reproductive, seed, and vegetative tissues from *I. cyaneum* from a previous study (Powell *et al.* 2022) and assembled the transcripts using the pipeline developed in Wheeler *et al.* (2022). Briefly, this pipeline first corrected read errors in the 128,433,717 raw reads using Rcorrector (Song and Florea 2015) and removed unfixable reads using *unfixable_filter.py* (Yang and Smith 2014) with Python v. 2.17.18 (van Rossum and de Boer 1991). We trimmed adaptor sequences from the filtered reads using Trimmomatic v. 0.39 (Bolger *et al.* 2014) and used the trimmed reads for *de novo* assembly with Trinity v. 2.11.0 (Grabherr *et al.* 2011). We removed apparent chimeric sequences using *run_chimera_detection.py* (Morales-Briones *et al.* 2021), with a reference BLAST database consisting of sequences from *Arabidopsis*, *Solanum*, and *Petunia*. We then used Corset v. 1.09 (Davidson and Oshlack 2014) to cluster transcripts and *filter_corset_output.py* (Yang and Smith 2014) to remove redundant transcripts. Finally, we predicted complete CDS from the Corset-filtered transcripts using TransDecoder v. 5.3.0 (Haas *et al.* 2013).

Next, we quantified gene expression by pseudo-aligning reads from each backcross individual to the predicted CDS in the transcriptome using Salmon (Patro *et al.* 2017). We calculated estimated read counts and TPM for each transcript. We imported Salmon quant files, partitioned by inferred *T*-locus genotype (*Tt/tt*), into DESeq2 with *tximport* (Soneson *et al.* 2015) and used the *DESeqDataSetFromTximport* function to create a DESeq analysis object, with treatments corresponding to the *T*-locus genotype. We quantified differential expression between these subsets using the *DESeq* function. We filtered the resulting transcripts by the adjusted p-value with a significance threshold of 0.05. This adjusted value accounts for multiple tests through the false discovery rate (FDR, Benjamini and Hochberg 1995); our cut-off corresponds to FDR of 5%.

In addition to examining differentially expressed genes between the phenotypic pools, we also computed pairwise correlations between the expression of *F3'H* and all other genes. We calculated Pearson correlation coefficients across the samples (n=24) and, to account for multiple comparisons, we adjusted the p-values using the Bonferroni correction.

Finally, as we expect that any regulator of *F3'H* may also regulate other flavonoid pathway genes, we used WGCNA v. 1.72.5 (Langfelder and Horvath 2008) to identify modules of co-expressed genes, focusing on modules that included *F3'H* and, in turn, are correlated with the color phenotype. WGCNA computes pairwise correlation coefficients, which then are converted to an adjacency matrix with the raw values raised to a soft-thresholding power (β) to approximate a scale-free network. For our data, we selected a β of 7, which corresponds to an R^2 value of 0.88 with the scale-free model and a mean connectivity of 20.4 (Supplementary Fig. 2). We initially used blockwise module detection on the full *de novo* transcriptomic dataset of 19,184 genes, and from this first pass, we retained modules with a correlation of 0.2 or greater with the trait of interest (color phenotype/inferred *T*-locus genotype). The filtered dataset contained 4854 genes, which allowed us to examine smaller modules (we set the minimum size to 20 genes). After hierarchical clustering, we merged modules that were 90% similar and re-calculated correlations between the module eigengenes and the trait. We exported the module containing *F3'H* to Cytoscape format using *exportNetworkToCytoscape* and extracted the topology overlap matrix (TOM) edge weights. We plotted the distribution of weights for edges containing *F3'H* and for all other edges and used Z-scores to capture how extreme each co-expression relationship is within the context of the module.

Phylogenetic analysis of *MYB12-like* genes and other SG7 MYBs

Our combined analyses of SNP association and gene expression strongly implicated an R2R3 MYB, which we refer to as *lochroma cyaneum MYB12-like* following the nomenclature in *Capsicum* (see Results). As R2R3 MYBs comprise a large group of functionally distinct transcription factors, we carried out phylogenetic analysis to identify the most closely related copies in other model Solanaceae. We used BLAST searches to retrieve the top hits from tomato, potato, groundcherry, chilipepper, *Nicotiana benthamiana*, and *Arabidopsis thaliana* (as an outgroup) and created a protein alignment with MAFFT v. 7 (Kato and Standley 2013) using default settings. As BLAST results suggested that the most similar sequences belonged to the flavonoid-regulating subgroup 7 (SG7) MYBs (see below), we included the R2 and R3 MYB domains through to the SG7 motif, which is widely conserved from *Arabidopsis* to tomato (Stracke *et al.* 2001; Stracke *et al.* 2007; Fernandez-Moreno *et al.* 2016). The downstream positions were trimmed as they were hypervariable and could not be confidently aligned. We estimated a maximum-likelihood phylogeny using this SG7 amino acid alignment with the best-

fitting model of amino acid substitutions (Q.plant+G4) and 1000 bootstrap replicates in IQ-TREE 2.3.6 (Nguyen *et al.* 2015; Minh *et al.* 2020).

Based on this broader phylogenetic analysis, we identified a small set of Solanaceae SG7 MYBs most closely related to the candidate locus. We next estimated a tree from full length coding sequences (CDS) from these closely related copies, which are more easily aligned. We included six additional *Lochrominae* sequences assembled by mapping reads from floral bud transcriptome data onto the *Lochroma cyaneum* genome sequence, again using STAR. Each of these six species is represented by two biological replicates; a consensus of the two was used for the phylogeny and the replicates were used to estimate the levels of *MYB12-like* expression in each species using Salmon as above. We estimated the maximum likelihood tree from the CDS alignment with IQ-TREE, using the best-fitting model of nucleotide substitutions (TIM3+F+G4).

Results

Localization of associated SNPs with flower color in the *Lochroma* genome

We recovered 92 SNPs that perfectly co-segregate with the two phenotypic pools, i.e., distinguish purple-flowered *Tt* and pink/red-flowered *tt* pools). The majority of these SNPs (49, 53%) fall on chromosome 5 of the *L. cyaneum* reference assembly. We also found 28 SNPs mapping to a roughly 620 Kb scaffold (00085) and the remainder (15) mapping to three additional unincorporated scaffolds (Supplementary Fig. 1). Our subsequent analyses suggest that these scaffolds represent segments of chromosome 5 that were not included during the assembly process (Powell *et al.* 2022). For example, scaffold00085 aligns well with tomato chromosome 5 (Supplementary Fig. 3), and 95% of the CDS retrieved from that scaffold have top hits on tomato chromosome 5. This region appears nested within the larger region of *L. cyaneum* chromosome 5 where most of the SNP associations are clustered (Fig. 2). The three smaller scaffolds with associated SNPs (Supplementary Fig. 1) also BLAST to tomato chromosome 5 and were also likely excluded during assembly. Thus, all SNPs recovered from the co-segregation analysis appear to be localized along a small region of *L. cyaneum* chromosome 5.

We carried out a case-control GWAS using the same set of variant calls. This analysis similarly retrieved associations exclusively on chromosome 5, with significant hits in the gene-

dense region in the last 500kb of the chromosome (Fig. 2 and Supplementary Fig. 4). This region of the genome contains 468 gene models (Supplementary Table 1), 352 of which are functionally annotated in the genome (Supplementary Table 2). Twenty-eight of these genes are annotated as transcription factors and only one corresponds to a known group of flavonoid regulators. BLAST searches suggested this locus (IC05g034110) is a member of the flavonol-regulating subgroup 7 MYBs (Stracke *et al.* 2007), with top hits to AtMYB111 and AtMYB12 in *A. thaliana* and to MYBs annotated as MYB12 and MYB-like in chilipepper and potato. We will refer to this gene as *lochroma cyaneum* MYB12-like (*IcMYB12-like*) based on the phylogenetic analysis (see below). The associated region also contains copies of one of the upstream pathway enzymes, chalcone synthase (CHS), as well as UDP-glycosyltransferase (UGT), which can glycosylate various flavonoids.

Patterns of differential expression and co-expression

Our DEseq2 analysis identified 58 significantly differentially expressed transcripts between the two phenotypic pools in our backcross (Supplementary Table 3). The MYB transcription factor *IcMYB12-like* appears as the sixth most strongly differentially expressed gene between the pools ($\log_2$ -fold change = -6.35, or ca. 82-fold lower expression in the pink/red pool). Its putative target, *F3'H*, is the eighth most differentially expressed ($\log_2$ -fold change = -5.83, or ca. 57-fold lower expression in the pink/red pool) and is tightly correlated to *MYB12-like* ($r=0.91$, $P<1.6\times 10^{-5}$, Fig. 3 and Supplementary Table 4). Note that the expression of *F3'H* in many of these individuals was previously measured with qPCR (Table 1; Smith and Rausher 2011); this analysis confirms the strength and degree of the differential expression between individuals presenting the alternate pigment phenotypes (Fig. 3a). A similar degree of differential expression was found for *FLS* between the two pools, and two other flavonoid pathway genes (*CHS* and *UGT*) also appear among the list of significantly DE genes (Supplementary Table 3). These patterns could indicate some degree of regulatory control of *IcMYB12-like* over other pathway steps.

In addition to examining DE genes between the two phenotypic pools, we explored co-expression of genes across the entire set of 24 backcross individuals. If *IcMYB12-like* indeed activates floral *F3'H* expression, we expect the two genes to show correlated expression and to belong to the same co-expressed module of genes. Consistent with this prediction, our WGCNA analysis recovered a module of 52 genes containing *F3'H* and *IcMYB12-like* (Supplementary Table 5). Out of the 34 modules found in the analysis, this module is the only one significantly

correlated with the pigment phenotype (purple vs. pink/red, $R^2 = -0.92$, $p = 1e^{-10}$, Supplementary Fig. 5). Within this module, *lcmYB12-like* is tightly co-expressed with *F3H* (Fig. 3c). The connectivity between *F3H* and *lcmYB12-like*, measured as topological overlap matrix (TOM) values from the WGCNA analysis, was the second highest in the set of all edges involving *F3H* (Z-score: 1.96) with only the edge connecting *F3H* and *FLS* having a higher value (Z-score: 2.56) (Supplementary Fig. 6). We also found a tight connection between *lcmYB12-like* and *FLS* (Z-score: 1.83), suggesting that both *F3H* and *FLS* are both regulated by *lcmYB12-like*. Three other flavonoid pathway genes, *CHS*, *CHI*, and *UGT* appear in the module associated with the phenotype (Supplementary Table 5), and all except for *CHI* are directly connected to *lcmYB12-like* (Fig. 3c). Eight other loci within phenotype-associated module are connected to *lcmYB12-like* (e.g., DETOX-35-like-2 and the F-box protein At5g07610-like), although no functional connection is known. Three of the genes connected to *lcmYB12-like* (R1A-10, the F-box protein At5g07610-like and *UGT*) fall in the same genomic region as *lcmYB12-like* (Supplementary Table 2), suggesting that these co-expression patterns may be related to co-localization within the genome (Michalak 2008). Indeed, differentially expressed transcripts are clustered around *lcmYB12-like* (Supplementary Fig. 7). Nevertheless, both *F3H* and *FLS* occur outside of the region containing *lcmYB12-like* (Fig. 2), excluding co-localization as an explanation for their strong co-expression with *lcmYB12-like*.

Phylogenetic relationship of *lcmYB12-like* to other MYB transcription factors

We used BLAST searches to retrieve similar sequences to *lcmYB12-like* (Supplementary Table 6). The top hits from *Arabidopsis* and Solanaceae genomes corresponded to members of subgroup 7 of R2R3 MYB transcription factors (Stracke *et al.* 2001). This subgroup controls flavonol production in *Arabidopsis* (Stracke *et al.* 2007) by regulating upstream steps such as *CHS*, *CHI*, and *FLS*, as well as the glycosyltransferases that stabilize these products. Maximum-likelihood analysis revealed that *lcmYB12-like* and highly similar sequences from pepper and potato are closely related to subgroup 7 but fall in a separate subclade, with strong support (Fig. 4a). Subgroup 7 MYBs have been well characterized in Solanaceae (e.g., Ballester *et al.* 2010; Song *et al.* 2019) and appear functionally similar to their orthologs in *Arabidopsis*. *lcmYB12-like* possesses an SG7 MYB that is closely related to these well-characterized MYB12 genes (IC05g030210, Fig. 4a) in addition to the divergent sequence (IC05g034110), which we refer to as a *MYB12-like* gene following the naming of *CaMYB12-like* in *Capsicum* (CA05g18430 in Fig. 4a).

Using additional BLAST searches beyond nightshade crops, we identified an additional member of the *MYB12-like* clade in the wolfberry genus *Lycium*, which we used to root the phylogeny including the additional *lochrominae* sequences (Fig. 4b). The resulting topology is similar to the species tree (Deanna *et al.* 2019) although many of the branches in *lochrominae* are unsupported given that the sequences present few differences (Supplementary Fig. 8). Examining the expression of *MYB12-like* in these taxa in relation to their floral flavonol production (Larter *et al.* 2019; Supplementary Table 7), we observed that species with higher *MYB12-like* expression also produce higher amounts of flavonols (Fig. 4b), which are mainly quercetin glycosides (Berardi *et al.* 2016). This pattern aligns well with the proposed function of *MYB12-like* in activating *F3'H*, which in turn produces DHQ, the precursor of quercetin (Fig. 1).

Discussion

This study aimed to identify the gene underlying the so-called *T*-locus, which acts as a transcriptional regulator of *F3'H* to determine flower color in the nightshade genus *lochroma*. By carrying out RNASeq of floral bud tissue from multiple backcross individuals with *T*-locus genotypes assigned based on flower color (Fig. 1b), we pinpointed an R2R3 MYB transcription factor as the strongest candidate for the *T*-locus. First, our SNP-association studies narrowed the candidate region to 10Mb near the end of chromosome 5 (Fig. 2 and Supplementary Fig. 4). This region of the genome contains 468 gene models, including 28 annotated as transcription factors. Among these, only one of these corresponds to a class of genes, SG7 MYBs, known to be involved in regulating flavonoid biosynthesis. This *MYB12-like* gene shows tightly correlated expression with *F3'H* across the backcross ($r=0.91$, $P=1.58 \times 10^{-05}$, Fig. 3a,b). Indeed, these two genes emerge as part of a compact module in transcriptome-wide co-expression analyses, with *F3'H* having a stronger connection with *MYB12-like* than any other gene in the floral transcriptome with the exception of *FLS* (Fig. 3b and Supplementary Fig. 6). Given that the effect of the *T*-locus could be due to coding sequences changes only, this set of analyses cannot conclusively eliminate other candidate transcription factors in the associated region of the genome. Nevertheless, our phylogenetic analyses identify *IcMYB12-like* as an ortholog of chillipepper *CaMYB12-like*, a recently characterized flavonoid regulator, which like its *lochroma* ortholog, acts as a positive regulator of *F3'H* (Wu *et al.* 2023). Together, these lines of evidence argue that the *T*-locus corresponds to the *MYB12-like* gene in *lochroma*, which drives the origin of red flowers by altering floral flavonoid composition. Below we discuss how these findings contribute to our broader understanding of flower color evolution.

Genetic basis of changes in floral hue

While genetic studies of flower color have long implicated subgroup 6 R2R3 MYBs as the major determinants of floral pigment intensity (e.g., Quattrocchio *et al.* 1999; Schwinn *et al.* 2006; Streisfeld *et al.* 2013; reviewed in Marin-Recinos and Pucker 2024), work on the genetic basis of changes in floral hue has implicated a wide variety of molecular mechanisms (Quattrocchio *et al.* 2006; Wessinger and Rausher 2012; Berardi *et al.* 2021). Differences in the type of anthocyanins produced, which in turn influence the type of flower color, can arise from shifts in gene regulation (either in *cis*- or *trans*-) as well as changes in the function of pathway enzymes (Smith and Rausher 2011; Hopkins and Rausher 2011; Smith *et al.* 2013; Wessinger and Rausher 2015; Wheeler *et al.* 2023). Nevertheless, the identity of transcription factors that influence the type of anthocyanin produced (as opposed to the overall amount) through changes in *F3'H* and/or *F3'5'H* expression has remained nebulous, even in well-studied model systems.

Because of the shared precursors within the flavonoid pathway, subgroup 7 (SG7) transcriptional regulators of flavonol production can directly influence anthocyanin production, and, as shown in the present study, the type of anthocyanin produced as well. The deeply conserved structure of the pathway presents multiple branching points where a single precursor can be converted in different products depending on the enzymes present and their properties (Winkel-Shirley 2001; Tohge *et al.* 2013). The colorful anthocyanins share dihydroflavonol precursors (DHK, DHQ, DHM) with flavonols, creating the potential for competition between DFR and FLS for these substrates (Fig. 1a). Thus, the upregulation of SG7 MYBs and, in turn, their targets (e.g., *CHS*, *CHI*, *F3H*, *FLS*) generally reduces anthocyanin production in favor of flavonols to produce paler flowers (Holton *et al.* 1993; Luo *et al.* 2016; Yuan *et al.* 2016; Wheeler *et al.* 2023). The precise effect of altering the expression of SG7 MYBs on flower color will, however, depend on their target genes and the substrate preferences of multifunctional pathway enzymes (e.g., DFR, FLS). The degree of competition for shared precursors may be mitigated by reduced spatiotemporal overlap in expression and/or functional specialization on different precursors (e.g., FLS on DHK and DFR on DHQ) (Choudhary and Pucker 2024).

In the case of *lochroma*, the ability of the SG7 *MYB12-like* gene to alter flower color is likely due to the combination of a narrowing of target genes and strong substrate preferences among downstream enzymes. While the pepper *CaMYB12-like* gene activates a broad suite of pathway genes (*CHS*, *CHI*, *F3H*, *F3'H*, *FLS*, *UGT*, Wu *et al.* 2023), the *lochroma* ortholog only shows strong co-expression with *F3'H* and *FLS* (plus weaker co-expression with *CHS* and

1 *UGT*), pointing to a reduced suite of targets, a functional shift that could be further explored with
 2 promoter-binding assays or transient transformation (e.g., Mehrrens *et al.* 2005; Yang *et al.*
 3 2023). The broad upstream action by *CaMYB12-like* is similar to that of the other well-known
 4 SG7 MYBs in Solanaceae (*SlMYB12* in tomato, Ballester *et al.* 2010; Fernandez-Moreno *et al.*
 5 2016; *NtMYB12* in tobacco, Song *et al.* 2019), suggesting that coordinated regulation of the first
 6 steps of the flavonoid pathway represents the ancestral state and that the functional shift toward
 7 specificity has occurred along the *MYB12-like* lineage leading to *lochroma*. Accordingly, the loss
 8 of *MYB12-like* expression in *I. gesnerioides* flowers is not associated with a complete disruption
 9 in floral flavonoid pigment production (Larter *et al.* 2019; Berardi *et al.* 2021), but a targeted
 10 reduction in DHQ through lower *F3H* expression. The resulting accumulation of DHK is not
 11 converted to kaempferol, likely because of coordinated loss of *FLS* expression and its low
 12 preference for DHQ, at least in the berry-fruited Solanaceae like *lochroma* (Bovy *et al.* 2007;
 13 Berardi *et al.* 2016; Rosa-Martínez *et al.* 2023). Instead, this DHK precursor is converted to red
 14 pelargonidin pigments by the DFR enzyme, which in *I. gesnerioides*, is specialized for DHK
 15 (Smith *et al.* 2013). Smith *et al.* (2013) hypothesized that, during the evolutionary transition from
 16 blue to red flowers, the *trans*-regulatory loss of *F3H* expression occurred first, allowing the flux
 17 to shift toward red pigmentation. Under this scenario, the selection would be expected to favor
 18 increased activity of DFR on DHK to allow efficient conversion to red pelargonidins.

19 This *MYB12-like*-mediated biochemical trade of blue anthocyanins plus flavonols for red
 20 anthocyanins alone may have also carried ecological consequences for relationships with
 21 pollinators. In addition to acting as co-pigments, flavonols increase floral UV-absorbance, which
 22 is attractive to moth pollinators (Sheehan *et al.* 2016), and can also enhance fly and bee
 23 visitation if associated with floral patterning (Koski and Ashman 2014). Indeed, insects comprise
 24 only 10% of pollinator visits to *lochroma gesnerioides* compared hummingbirds, which account
 25 for 90% of visits (Smith *et al.* 2008). This lack of UV-absorbing flavonols is isolated to *I.*
 26 *gesnerioides* flowers as the leaves produce comparable amounts of flavonols (specifically
 27 quercetin) as the blue-flowered *I. cyaneum* (Berardi *et al.* 2016) and the expression of *F3H* is
 28 actually higher in *I. gesnerioides* leaves than in those of *I. cyaneum* (Smith and Rausher 2011).
 29 The targeted effects of *MYB12-like* on floral flavonols may have thus created an accessible
 30 evolutionary pathway to red flowers, given that a loss of quercetin across the entire plant would
 31 carry significant negative pleiotropic effects (Ryan *et al.* 2001; Singh *et al.* 2021).

32
 33 MYB transcription factors in the evolution of species differences

Closely related species of flowering plants are often distinguished by subtle differences in their reproductive organs, e.g., in the color, shape, scent, or pubescence of flowers or fruits. MYB transcription factors control many of these aspects of morphological development and epidermal cell fate (Ramsay and Glover 2005; Hileman 2014), which may help to explain their prevalence in underlying fixed differences between species (e.g., Preston *et al.* 2011; Gates *et al.* 2018; Castillejo *et al.* 2020; Yarahmadov *et al.* 2020). In fact, MYB transcription factors may act as speciation genes when the phenotypic differences resulting from changes in their function or expression leads to reproductive isolation (Streisfeld *et al.* 2013; Sheehan *et al.* 2016; Lüthi *et al.* 2022). Through its simultaneous effects on visible anthocyanins and UV-absorbing pigments, changes in floral *MYB12-like* expression could have played a role in species divergence, although the split between the red-flowered clade containing *I. gesnerioides* and its blue-flowered relatives likely occurred 5 to 10 million years ago (Huang *et al.* 2023), and the two lineages no longer occur in hybrid zones. The *I. arborescens* complex (the “A” clade sensu Smith and Baum (2006)) presents a stronger opportunity for dissecting the role of *MYB12-like* in floral isolation as red-flowered, low-flavonol primarily-hummingbird-pollinated species (e.g., *I. edule*) co-occur and hybridize with flavonol-rich insect-pollinated species (e.g., *I. arborescens*) (Smith *et al.* 2008; Fig. 4b).

Cis-regulatory mutations involving MYBs appear to be a major target for evolutionary transitions, and our results suggest that regulatory changes, as opposed to functional variation, drive the effects of *MYB12-like* on flower color in *lochroma*. First, the *MYB12-like* sequence from *I. gesnerioides* shows 6 fixed amino acid differences from its closest blue-flowered relative (*I. calycinum*), however all but one of these variants (a threonine indel close to the 3' end, Supplementary Fig. 8) are segregating across *lochroma* species with both low and high flavonol accumulation (Fig. 4b). Moreover, *MYB12-like* expression levels are strongly predictive of pathway activity. Within the backcross, the red *I. gesnerioides* parent allele of *MYB12-like* is expressed at extremely low levels, which in homozygous state translates to a near absence of *F3'H* expression (Fig. 3a). This relationship extends above the species level, where lower levels of *MYB12-like* expression appear to be associated with lower levels of quercetin flavonols (Fig. 4b). Beyond *lochroma*, *cis*-regulatory mutations at MYB transcription factors frequently contribute to within and among species differences in flower color (Streisfeld *et al.* 2013; Martins *et al.* 2017; Fattorini and Ó'Maoiléidigh 2022), a pattern that has been attributed to their comparatively limited pleiotropic effects (Sobel and Streisfeld 2013). Nevertheless, the precise changes in the MYB promoters are unknown in these natural systems. Identifying the causal mutation(s) will require fine dissection of the promoter region along with *in vivo* or *in vitro* assays

of various constructs (e.g., Espley *et al.* 2009; Jia *et al.* 2021). Although transformation remains challenging outside of model systems, pinpointing these causal variants is important for ultimately understanding how and why MYBs and the modules they control can be deployed in new developmental contexts.

Conclusions

As a powerful group of antioxidants, flavonols have long been the focus of efforts in plant breeding, resulting in a detailed understanding of the subgroup 7 MYBs that largely control their expression across flowering plants. Within the nightshades, the best known of these MYBs are the orthologs of MYB12, which contribute to stress tolerance in tobacco (Song *et al.* 2019) and the color of the fruit peel in tomato (Ballester *et al.* 2010) via their effects on flavonoid production. This gene family is also expressed in Solanaceae flowers (Zheng *et al.* 2021), activating multiple branches of the pathway to provide both flavonol co-pigments and the substrates for anthocyanin biosynthesis. Our work reveals that, in addition to this canonical 'MYB12' group of SG7 MYBs, *Ipomoea* flowers also express a more divergent 'MYB12-like' lineage that has evolved narrow specificity for *FLS* and *F3'H*. This specialization, together with flower-specific expression, allows *IcMYB12-like* to act as the switch between blue and red flowers. Piecing together the origin of this gene's role in floral flavonoid production will require targeted comparative genomic, transcriptomic, and metabolomic studies to pinpoint the timing of this functional shift within the nightshades, as well as additional experimental work to identify the upstream regulators of flavonoid MYBs and their binding motifs within *cis*-regulatory regions (e.g., Chopy *et al.* 2024). Overall, our study underscores the importance of studying flower color evolution in the context of the entire flavonoid pathway as its conserved branching architecture shapes trade-offs between the visible pigments and 'colorless' compounds.

Data Availability Statement

RNA-seq data from the backcross individuals have been uploaded to the SRA under Bioproject PRJNA1092111. Data for the other six species have been uploaded to SRA Bioproject PRJNA1102413. Code and additional data used in the analyses are available at DOI 10.17605/OSF.IO/J5M8F. Code for *de novo* transcriptome assembly are available through the repository associated with Wheeler *et al.* (2022), <https://osf.io/b7gcp/>.

Acknowledgments

The authors thank Adrian Powell (Boyce Thompson Institute) for providing access to the *lochroma cyaneum* genome through the SolGenomics website. We also thank two anonymous reviewers for their constructive feedback.

Funding

This work was supported by the National Science Foundation [grant number DEB 1553114 to S.D.S.]. The RMACC Summit and Alpine supercomputers used for the computational work are also supported by the National Science Foundation [grant numbers ACI-1532235, ACI-1532236, and ACI 2201538].

Conflict of Interest

No conflict of interest declared.

References

- Albert NW, Davies KM, Lewis DH, Zhang H, Montefiori M, Brendolise C, Boase MR, Ngo H, Jameson PE, Schwinn KE. 2014. A conserved network of transcriptional activators and repressors regulates anthocyanin pigmentation in eudicots. *Plant Cell*. 26(3):962-980, doi:10.1105/tpc.113.122069.
- Albert NW, Lewis DH, Zhang H, Schwinn KE, Jameson PE, Davies KM. 2011. Members of an R2R3-MYB transcription factor family in petunia are developmentally and environmentally regulated to control complex floral and vegetative pigmentation patterning. *Plant J*. 65(5):771-784, doi:10.1111/j.1365-313X.2010.04465.x.
- Auge GA, Penfield S, Donohue K. 2019. Pleiotropy in developmental regulation by flowering-pathway genes: Is it an evolutionary constraint? *New Phytol*. 224(1):55-70, doi:10.1111/nph.15901.
- Ballester AR, Molthoff J, de Vos R, Hekkert B, Orzaez D, Fernández-Moreno JP, Tripodi P, Grandillo S, Martin C, Heldens J et al. 2010. Biochemical and molecular analysis of pink tomatoes: Deregulated expression of the gene encoding transcription factor *SlMYB12* leads to pink tomato fruit color. *Plant Physiol*. 152(1):71-84, doi:10.1104/pp.109.147322.
- Benjamini Y, Hochberg Y. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc B*. 57(1): 289-300, doi: 10.1111/j.2517-6161.1995.tb02031.x
- Berardi AE, Esfeld K, Jäggi L, Mandel T, Cannarozzi GM, Kuhlemeier C. 2021. Complex evolution of novel red floral color in petunia. *Plant Cell*. 33(7):2273-2295, doi:10.1093/plcell/koab114.
- Berardi AE, Hildreth SB, Helm RF, Winkel BS, Smith SD. 2016. Evolutionary correlations in flavonoid production across flowers and leaves in the lochrominae (Solanaceae). *Phytochemistry*. 130:119-127, doi:10.1016/j.phytochem.2016.05.007.
- Bolger AM, Lohse M, Usadel B. 2014. Trimmomatic: A flexible trimmer for illumina sequence data. *Bioinformatics*. 30(15):2114-2120, doi:10.1093/bioinformatics/btu170.
- Bow A, Schijlen E, Hall RD. 2007. Metabolic engineering of flavonoids in tomato (*Solanum lycopersicum*): The potential for metabolomics. *Metabolomics*. 3:399-412, doi:10.1007/s11306-007-0074-2.
- Byers KJ, Vela JP, Peng F, Riffell JA, Bradshaw HD. 2014. Floral volatile alleles can contribute to pollinator-mediated reproductive isolation in monkeyflowers (*Mimulus*). *Plant J*. 80(6):1031-1042, doi:10.1111/tpj.12702.
- Cabanettes F, Klopp C. 2018. D-genies: Dot plot large genomes in an interactive, efficient and simple way. *PeerJ*. 6:e4958, doi:10.7717/peerj.4958.
- Casbon J. 2012. Pyvcf - a variant call format parser for python v. 0.3.0. ed. <https://github.com/jamescasbon/PyVCF>.
- Castillejo C, Waurich V, Wagner H, Ramos R, Oiza N, Muñoz P, Triviño JC, Caruana J, Liu Z, Cobo N et al. 2020. Allelic variation of MYB10 is the major force controlling natural variation in skin and flesh color in strawberry (*Fragaria* spp.) fruit. *Plant Cell*. 32(12):3723-3749, doi:10.1105/tpc.20.00474.
- Chopy M, Cavallini-Speisser Q, Chambrier P, Morel P, Just J, Hugouvieu V, Rodrigues Bento S, Zubieta C, Vandenbussche M, Monniaux M. 2024. Cell layer-specific expression of the homeotic MADS-box transcription factor *PhDEF* contributes to modular petal morphogenesis in petunia. *Plant Cell*. 36(2): 324-345, doi:10.1093/plcell/koad258.
- Choudhary N, Pucker B. 2024. Conserved amino acid residues and gene expression patterns associated with the substrate preferences of the competing enzymes FLS and DFR. *PLoS One*. 19(8): e0305837, doi: 10.1371/journal.pone.0305837
- Danecek P, Bonfield JK, Liddle J, Marshall J, Ohan V, Pollard MO, Whitwham A, Keane T, McCarthy SA, Davies RM, Li H. 2021. Twelve years of SAMtools and BCFtools. *Gigascience*. 10(2): giab008, doi: 10.1093/gigascience/giab008.
- Davidson NM, Oshlack A. 2014. Corset: Enabling differential gene expression analysis for de novo assembled transcriptomes. *Genome Biol*. 15(7):410, doi:10.1186/s13059-014-0410-6.
- Deanna R, Larter MD, Barboza GE, Smith SD. 2019. Repeated evolution of a morphological novelty: A phylogenetic analysis of the inflated fruiting calyx in the Physalideae tribe (Solanaceae). *Am J Bot*. 106(2):270-279, doi:10.1002/ajb2.1242.
- Deanna R, González Ramírez I, Särkinen T, Knapp S, Sauquet H, Smith SD. 2025. Late Cretaceous origins for major nightshade lineages from total evidence timetree analysis. *BioRxiv* doi:10.1101/2025.07.22.666174

- 1 Des Marais DL, Rausher MD. 2010. Parallel evolution at multiple levels in the origin of hummingbird
2 pollinated flowers in *Ipomoea*. *Evolution*. 64(7):2044-2054, doi:10.1111/j.1558-
3 5646.2010.00972.x.
- 4 de Vetten N, ter Horst J, van Schaik HP, de Boer A, Mol J, Koes R. 1999. A cytochrome b5 is required for
5 full activity of flavonoid 3', 5'-hydroxylase, a cytochrome P450 involved in the formation of blue
6 flower colors. *Proc. Natl. Acad. Sci. U.S.A.* 96(2): 778-783, doi: 10.1073/pnas.96.2.778.
- 7 Dobin A, Davis CA, Schlesinger F, Drenkow J, Zaleski C, Jha S, Batut P, Chaisson M, Gingeras TR.
8 2013. Star: Ultrafast universal rna-seq aligner. *Bioinformatics*. 29(1):15-21,
9 doi:10.1093/bioinformatics/bts635.
- 10 Dubos C, Stracke R, Grotewold E, Weisshaar B, Martin C, Lepiniec L. 2010. MYB transcription factors in
11 *Arabidopsis*. *Trends Plant Sci.* 15(10):573-581, doi:10.1016/j.tplants.2010.06.005.
- 12 Espley RV, Brendolise C, Chagné D, Kutty-Amma S, Green S, Volz R, Putterill J, Schouten HJ, Gardiner
13 SE, Hellens RP et al. 2009. Multiple repeats of a promoter segment causes transcription factor
14 autoregulation in red apples. *Plant Cell*. 21(1):168-183, doi:10.1105/tpc.108.059329.
- 15 Fattorini R, Ó'Maoiléidigh DS. 2022. *Cis*-regulatory variation expands the colour palette of the
16 Brassicaceae. *J Exp Bot.* 73(19): 6511-6515, doi: 10.1093/jxb/erac366.
- 17 Feller A, Machemer K, Braun EL, Grotewold E. 2011. Evolutionary and comparative analysis of myb and
18 bhlh plant transcription factors. *Plant J.* 66(1):94-116, doi:10.1111/j.1365-313X.2010.04459.x.
- 19 Fernandez-Moreno JP, Tzfadia O, Forment J, Presa S, Rogachev I, Meir S, Orzaez D, Aharoni A, Granell
20 A. 2016. Characterization of a new pink-fruited tomato mutant results in the identification of a null
21 allele of the *S/MYB12* transcription factor. *Plant Physiol.* 171(3):1821-1836,
22 doi:10.1104/pp.16.00282.
- 23 Gates DJ, Olson BJSC, Clemente TE, Smith SD. 2018. A novel R3 MYB transcriptional repressor
24 associated with the loss of floral pigmentation in *Ichroma*. *New Phytol.* 217(3):1346-1356,
25 doi:10.1111/nph.14830.
- 26 Gates DJ, Strickler SR, Mueller LA, Olson BJ, Smith SD. 2016. Diversification of R2R3-MYB transcription
27 factors in the tomato family Solanaceae. *J Mol Evol.* 83(1-2):26-37, doi:10.1007/s00239-016-
28 9750-z.
- 29 Grabherr MG, Haas BJ, Yassour M, Levin JZ, Thompson DA, Amit I, Adiconis X, Fan L, Raychowdhury R,
30 Zeng Q et al. 2011. Full-length transcriptome assembly from RNA-seq data without a reference
31 genome. *Nat Biotech.* 29(7):644-652, doi:10.1038/nbt.1883.
- 32 Gonzalez A, Mendenhall J, Huo Y, Lloyd A. 2009. TTG1 complex MYBs, MYB5 and TT2, control outer
33 seed coat differentiation. *Dev Biol.* 325(2): 412-421, doi: 10.1016/j.ydbio.2008.10.005.
- 34 Haas BJ, Papanicolaou A, Yassour M, Grabherr M, Blood PD, Bowden J, Couger MB, Eccles D, Li B,
35 Lieber M et al. 2013. De novo transcript sequence reconstruction from RNA-seq using the Trinity
36 platform for reference generation and analysis. *Nat Protoc.* 8(8):1494-1512, pp. 1494-512,
37 doi:10.1038/nprot.2013.084.
- 38 Hileman LC. 2014. Trends in flower symmetry evolution revealed through phylogenetic and
39 developmental genetic advances. *Phil Trans R Soc B.* 369(1648), doi:10.1098/rstb.2013.0348.
- 40 Holton TA, Brugliera F, Tanaka Y. 1993. Cloning and expression of flavonol synthase from *Petunia*
41 *hybrida*. *Plant J.* 4(6):1003-1010, doi:10.1046/j.1365-313x.1993.04061003.x.
- 42 Hopkins R, Rausher MD. 2011. Identification of two genes causing reinforcement in the texas wildflower
43 *Phlox drummondii*. *Nature.* 469(7330):411-414, doi:10.1038/nature09641.
- 44 Huang J, Xu W, Zhai J, Hu Y, Guo J, Zhang C, Zhao Y, Zhang L, Martine C, Ma H et al. 2023. Nuclear
45 phylogeny and insights into whole-genome duplications and reproductive development of
46 Solanaceae plants. *Plant Comm.* 4(4):100595, doi:10.1016/j.xplc.2023.100595.
- 47 Jia D, Wu P, Shen F, Li W, Zheng X, Wang Y, Yuan Y, Zhang X, Han Z. 2021. Genetic variation in the
48 promoter of an R2R3-MYB transcription factor determines fruit malate content in apple (*malus*
49 *domestica* borkh.). *Plant Physiol.* 186(1):549-568, doi:10.1093/plphys/kiab098.
- 50 Johnson ET, Yi H, Shin B, Oh BJ, Cheong H, Choi G. 1999. *Cymbidium hybrida* dihydroflavonol 4-
51 reductase does not efficiently reduce dihydrokaempferol to produce orange pelargonidin-type
52 anthocyanins. *Plant J.* 19(1): 81-85, doi: 10.1046/j.1365-313x.1999.00502.
- 53 Johnson ET, Ryu S, Yi H, Shin B, Cheong H, Choi G. 2001. Alteration of a single amino acid changes the
54 substrate specificity of dihydroflavonol 4-reductase. *Plant J.* 25(3): 325-333, doi: 10.1046/j.1365-
55 313x.2001.00962.

- 1 Katoh K, Standley DM. 2013. MAFFT multiple sequence alignment software version 7: Improvements in
2 performance and usability. *Mol Biol Evol.* 30(4):772-780, doi:10.1093/molbev/mst010.
- 3 Kimura S, Koenig D, Kang J, Yoong FY, Sinha N. 2008. Natural variation in leaf morphology results from
4 mutation of a novel *knox* gene. *Curr Biol.* 18(9):672-677, doi:10.1016/j.cub.2008.04.008.
- 5 Koski MH, Ashman T-L. 2014. Dissecting pollinator responses to a ubiquitous ultraviolet floral pattern in
6 the wild. *Funct. Ecol.* 28(4):868-877, doi:https://doi.org/10.1111/1365-2435.12242.
- 7 Langfelder P, Horvath S. 2008. Wgcna: An R package for weighted correlation network analysis. *BMC*
8 *Bioinformatics.* 9:559, doi:10.1186/1471-2105-9-559.
- 9 Larter M, Dunbar-Wallis A, Berardi AE, Smith SD. 2018. Convergent evolution at the pathway level:
10 Predictable regulatory changes during flower color transitions. *Mol Biol Evol.* 35(9):2159-2169,
11 doi:10.1093/molbev/msy117.
- 12 Larter M, Dunbar-Wallis A, Berardi AE, Smith SD. 2019. Developmental control of convergent floral
13 pigmentation across evolutionary timescales. *Dev Dyn.* 248(11):1091-1100, doi:10.1002/dvdy.82.
- 14 Luo P, Ning G, Wang Z, Shen Y, Jin H, Li P, Huang S, Zhao J, Bao M. 2016. Disequilibrium of flavonol
15 synthase and dihydroflavonol-4-reductase expression associated tightly to white vs. red color
16 flower formation in plants. *Front Plant Sci.* 6: 1257 doi: 10.3389/fpls.2015.01257.
- 17 Lüthi MN, Berardi AE, Mandel T, Freitas LB, Kuhlemeier C. 2022. Single gene mutation in a plant myb
18 transcription factor causes a major shift in pollinator preference. *Curr Biol.* 32(24):5295-
19 5308.e5295, doi:10.1016/j.cub.2022.11.006.
- 20 Lynch VJ, Wagner GP. 2008. Resurrecting the role of transcription factor change in developmental
21 evolution. *Evolution.* 62(9):2131-2154, doi:10.1111/j.1558-5646.2008.00440.x.
- 22 Marin-Recinos MF, Pucker B. 2024. Genetic factors explaining anthocyanin pigmentation differences.
23 *BMC Plant Biol.* 24: 627, doi:10.1186/s12870-024-05316-w.
- 24 Martins TR, Jiang P, Rausher MD. 2017. How petals change their spots: *Cis*-regulatory re-wiring in
25 *Clarkia* (Onagraceae). *New Phytol.* 216(2):510-518, doi:10.1111/nph.14163.
- 26 Marçais G, Delcher AL, Phillippy AM, Coston R, Salzberg SL, Zimin A. 2018. Mummer4: A fast and
27 versatile genome alignment system. *PLoS Comp Biol.* 14(1):e1005944,
28 doi:10.1371/journal.pcbi.1005944.
- 29 Mehrtens F, Kranz H, Bednarek P, Weisshaar B. 2005. The *Arabidopsis* transcription factor MYB12 is a
30 flavonol-specific regulator of phenylpropanoid biosynthesis. *Plant Physiol.* 138:1083-1096, doi:
31 10.1104/pp.104.058032.
- 32 Michalak P. 2008. Coexpression, coregulation, and cofunctionality of neighboring genes in eukaryotic
33 genomes. *Genomics.* 91(3):243-248, doi:10.1016/j.ygeno.2007.11.002.
- 34 Millar AA, Gubler F. 2005. The *Arabidopsis* GAMYB-like genes, MYB33 and MYB65, are microRNA-
35 regulated genes that redundantly facilitate anther development. *Plant Cell.* 17(3): 705-721, doi:
36 10.1105/tpc.104.027920.
- 37 Minh BQ, Schmidt HA, Chernomor O, Schrempf D, Woodhams MD, von Haeseler A, Lanfear R. 2020. Iq-
38 tree 2: New models and efficient methods for phylogenetic inference in the genomic era. *Mol Biol*
39 *Evol.* 37(5):1530-1534, doi:10.1093/molbev/msaa015.
- 40 Mol J, Grotewold E, Koes R. 1998. How genes paint flowers and seeds. *Trends Plant Sci.* 3(6):212-217,
41 doi:10.1016/S1360-1385(98)01242-4.
- 42 Morales-Briones DF, Kadereit G, Tefarikis DT, Moore MJ, Smith SA, Brockington SF, Timoneda A, Yim
43 WC, Cushman JC, Yang Y. 2021. Disentangling sources of gene tree discordance in
44 phylogenomic data sets: Testing ancient hybridizations in Amaranthaceae S.L. *Syst Biol.*
45 70(2):219-235, doi:10.1093/sysbio/syaa066.
- 46 Mu Q, Wei J, Longest HK, Liu H, Char SN, Hinrichsen JT, Tibbs-Cortes LE, Schoenbaum GR, Yang B, Li
47 X et al. 2024. A MYB transcription factor underlying plant height in sorghum qht7.1 and maize
48 brachytic 1 loci. *Plant J.* 120(5):2172-2192, doi:10.1111/tpj.17111.
- 49 Nguyen LT, Schmidt HA, von Haeseler A, Minh BQ. 2015. Iq-tree: A fast and effective stochastic
50 algorithm for estimating maximum-likelihood phylogenies. *Mol Biol Evol.* 32(1):268-274,
51 doi:10.1093/molbev/msu300.
- 52 Panchy N, Lehti-Shiu M, Shiu SH. 2016. Evolution of gene duplication in plants. *Plant Physiol.*
53 171(4):2294-2316, doi:10.1104/pp.16.00523.
- 54 Patro R, Duggal G, Love MI, Irizarry RA, Kingsford C. 2017. Salmon provides fast and bias-aware
55 quantification of transcript expression. *Nat Methods.* 14(4):417-419, doi:10.1038/nmeth.4197.

- 1 Pollak PE, Vogt T, Mo Y, Taylor LP. 1993. Chalcone synthase and flavonol accumulation in stigmas and
2 anthers of *Petunia hybrida*. Plant Physiol. 102: 925-932, doi:10.1104/pp.102.3.925.
- 3 Powell AF, Zhang J, Hauser D, Vilela JA, Hu A, Gates DJ, Mueller LA, Li FW, Strickler SR, Smith SD.
4 2022. Genome sequence for the blue-flowered andean shrub *Lochroma cyaneum* reveals
5 extensive discordance across the berry clade of solanaceae. Plant Genome. 15(3):e20223,
6 doi:10.1002/tpg2.20223.
- 7 Preston JC, Martinez CC, Hileman LC. 2011. Gradual disintegration of the floral symmetry gene network
8 is implicated in the evolution of a wind-pollination syndrome. Proc. Natl. Acad. Sci. U.S.A.
9 108(6):2343-2348, doi:10.1073/pnas.1011361108.
- 10 Prud'homme B, Gompel N, Rokas A, Kassner VA, Williams TM, Yeh SD, True JR, Carroll SB. 2006.
11 Repeated morphological evolution through cis-regulatory changes in a pleiotropic gene. Nature.
12 440(7087):1050-1053, doi:10.1038/nature04597.
- 13 Quattrocchio F, Verweij W, Kroon A, Spelt C, Mol J, Koes R. 2006. Ph4 of petunia is an R2R3 MYB
14 protein that activates vacuolar acidification through interactions with basic-helix-loop-helix
15 transcription factors of the anthocyanin pathway. Plant Cell. 18(5):1274-1291,
16 doi:10.1105/tpc.105.034041.
- 17 Quattrocchio F, Wing J, van der Woude K, Souer E, de Vetten N, Mol J, Koes R. 1999. Molecular analysis
18 of the anthocyanin2 gene of petunia and its role in the evolution of flower color. Plant Cell.
19 11(8):1433-1444, doi:10.1105/tpc.11.8.1433.
- 20 Ramsay NA, Glover BJ. 2005. MYB-bHLH-WD40 protein complex and the evolution of cellular diversity.
21 Trends Plant Sci. 10(2):63-70, doi:10.1016/j.tplants.2004.12.011.
- 22 Rausher MD. 2006. The evolution of flavonoids and their genes. In: Grotewold E, editor. The science of
23 flavonoids. New York, NY: Springer New York. p. 175-211, doi:10.1007/978-0-387-28822-2_7.
- 24 Rosa-Martínez E, Bow A, Plazas M, Tikunov Y, Prohens J, Pereira-Dias L. 2023. Genetics and breeding
25 of phenolic content in tomato, eggplant and pepper fruits. Front Plant Sci. 14:1135237,
26 doi:10.3389/fpls.2023.1135237.
- 27 Ryan KG, Swinny EE, Winefield C, Markham KR. 2001. Flavonoids and UV photoprotection in
28 *Arabidopsis* mutants. Z. Naturforsch., C: Biosci. 56(9-10):745-754, doi:10.1515/znc-2001-9-1013.
- 29 Sagawa JM, Stanley LE, LaFountain AM, Frank HA, Liu C, Yuan YW. 2016. An R2R3 MYB transcription
30 factor regulates carotenoid pigmentation in *Mimulus lewisii* flowers. New Phytol. 209(3):1049-
31 1057, doi:10.1111/nph.13647.
- 32 Schwinn K, Venail J, Shang Y, Mackay S, Alm V, Butelli E, Oyama R, Bailey P, Davies K, Martin C. 2006.
33 A small family of MYB-regulatory genes controls floral pigmentation intensity and patterning in the
34 genus antirrhinum. Plant Cell. 18(4):831-851, doi:10.1105/tpc.105.039255.
- 35 Schwinn KE, Boase MR, Bradley JM, Lewis DH, Deroles SC, Martin CR, Davies KM. 2014. MYB and
36 bHLH transcription factor transgenes increase anthocyanin pigmentation in *Petunia* and
37 *Lisianthus* plants, and the *Petunia* phenotypes are strongly enhanced under field conditions.
38 Front Plant Sci. 5:603, doi:10.3389/fpls.2014.00603.
- 39 Sheehan H, Moser M, Klahre U, Esfeld K, Dell'Olio A, Mandel T, Metzger S, Vandenbussche M, Freitas
40 L, Kuhlmeier C. 2016. MYB-FL controls gain and loss of floral UV absorbance, a key trait
41 affecting pollinator preference and reproductive isolation. Nat Genet. 48(2):159-166,
42 doi:10.1038/ng.3462.
- 43 Shin J, Park E, Choi G. 2007. PIF3 regulates anthocyanin biosynthesis in an HY5-dependent manner with
44 both factors directly binding anthocyanin biosynthetic gene promoters in *Arabidopsis*. Plant J.
45 49(6): 981-994, doi: 10.1111/j.1365-3113X.2006.03021.
- 46 Shiu SH, Shih MC, Li WH. 2005. Transcription factor families have much higher expansion rates in plants
47 than in animals. Plant Physiol. 139(1):18-26, doi:10.1104/pp.105.065110.
- 48 Singh P, Arif Y, Bajguz A, Hayat S. 2021. The role of quercetin in plants. Plant Physiol Biochem. 166:10-
49 19, doi:10.1016/j.plaphy.2021.05.023.
- 50 Smith SD, Baum DA. 2006. Phylogenetics of the florally diverse Andean clade lochrominae (solanaceae).
51 Am J Bot. 93(8):1140-1153, doi:10.3732/ajb.93.8.1140.
- 52 Smith SD, Hall SJ, Izquierdo PR, Baum DA. 2008. Comparative pollination biology of sympatric and
53 allopatric andean lochroma (Solanaceae). Ann Mo Bot Gard. 95(4):600-617, doi:
54 10.3417/2007037.
- 55 Smith SD, Rausher MD. 2011. Gene loss and parallel evolution contribute to species difference in flower
56 color. Mol Biol Evol. 28(10):2799-2810, doi:10.1093/molbev/msr109.

- 1 Smith SD, Wang S, Rausher MD. 2013. Functional evolution of an anthocyanin pathway enzyme during a
2 flower color transition. *Mol Biol Evol.* 30(3):602-612, doi:10.1093/molbev/mss255.
- 3 Sobel JM, Streisfeld MA. 2013. Flower color as a model system for studies of plant evo-devo. *Front Plant*
4 *Sci.* 4:321, doi:10.3389/fpls.2013.00321.
- 5 Sonesson C, Love MI, Robinson MD. 2015. Differential analyses for RNA-seq: Transcript-level estimates
6 improve gene-level inferences. *F1000Research.* 4:1521, doi:10.12688/f1000research.7563.2.
- 7 Song L, Florea L. 2015. Rcorrector: Efficient and accurate error correction for illumina rna-seq reads.
8 *Gigascience.* 4:48, doi:10.1186/s13742-015-0089-y.
- 9 Song Z, Luo Y, Wang W, Fan N, Wang D, Yang C, Jia H. 2019. NtMYB12 positively regulates flavonol
10 biosynthesis and enhances tolerance to low pi stress in *Nicotiana tabacum*. *Front Plant Sci.*
11 10:1683, doi:10.3389/fpls.2019.01683.
- 12 Stracke R, Favory JJ, Gruber H, Bartelniewoehner L, Bartels S, Binkert M, Funk M, Weisshaar B, Ulm R.
13 2010. The *Arabidopsis* bZIP transcription factor *HY5* regulates expression of the PFG1/MYB12
14 gene in response to light and ultraviolet-B radiation. *Plant Cell Environ.* 33(1):88-103, doi:
15 10.1111/j.1365-3040.2009.02061.
- 16 Stracke R, Ishihara H, Hup G, Barsch A, Mehrrens F, Niehaus K, Weisshaar B. 2007. Differential
17 regulation of closely related R2R3 MYB transcription factors controls flavonol accumulation in
18 different parts of the arabidopsis thaliana seedling. *Plant J.* 50(4):660-677, doi:10.1111/j.1365-
19 313X.2007.03078.
- 20 Stracke R, Werber M, Weisshaar B. 2001. The R2R3 MYB gene family in *Arabidopsis thaliana*. *Curr Opin*
21 *Plant Biol.* 4(5):447-456, doi:10.1016/s1369-5266(00)00199-0.
- 22 Streisfeld MA, Young WN, Sobel JM. 2013. Divergent selection drives genetic differentiation in an r2r3-
23 myb transcription factor that contributes to incipient speciation in *Mimulus aurantiacus*. *PLoS*
24 *Genet.* 9(3):e1003385, doi:10.1371/journal.pgen.1003385.
- 25 Todesco M, Bercovich N, Kim A, Imerovski I, Owens GL, Dorado Ruiz Ó, Holalu SV, Madilao LL, Jahani
26 M, Légaré JS et al. 2022. Genetic basis and dual adaptive role of floral pigmentation in
27 sunflowers. *Elife.* 11:e72072, doi:10.7554/eLife.72072.
- 28 Tohge T, Watanabe M, Hoefgen R, Fernie AR. 2013. The evolution of phenylpropanoid metabolism in the
29 green lineage. *Crit Rev Biochem Mol Biol.* 48(2):123-152, doi:10.3109/10409238.2012.758083.
- 30 van Rossum G, de Boer J. 1991. Interactively testing remote servers using the python programming
31 language. *CWI Quarterly* 4: 283–303.
- 32 Wang YS, Xu YJ, Gao LP, Yu O, Wang XZ, He XJ, Jiang XL, Liu YJ, Xia T. 2014. Functional analysis of
33 flavonoid 3',5'-hydroxylase from tea plant (*Camellia sinensis*): critical role in the accumulation of
34 catechins. *BMC Plant Biol.* 14: 347, doi: 10.1186/s12870-014-0347-7.
- 35 Wessinger CA, Rausher MD. 2012. Lessons from flower colour evolution on targets of selection. *J Exp*
36 *Botany.* 63(16):5741-5749, doi:10.1093/jxb/ers267.
- 37 Wessinger CA, Rausher MD. 2015. Ecological transition predictably associated with gene degeneration.
38 *Mol Biol Evol.* 32(2):347-354, doi:10.1093/molbev/msu298.
- 39 Wheeler LC, Wing BA, Smith SD. 2020. Structure and contingency determine mutational hotspots for
40 flower color evolution. *Evol Lett.* 26: 61-74, doi: 10.1002/evl3.212.
- 41 Wheeler LC, Dunbar-Wallis A, Schutz K, Smith SD. 2023. Evolutionary walks through flower colour space
42 driven by gene expression in *Petunia* and allies (Petunieae). *Proc R Soc B.* 290(2002):20230275,
43 doi:10.1098/rspb.2023.0275.
- 44 Winkel-Shirley B. 2001. Flavonoid biosynthesis. A colorful model for genetics, biochemistry, cell biology,
45 and biotechnology. *Plant Physiol.* 126(2):485-493, doi:10.1104/pp.126.2.485.
- 46 Wray GA. 2007. The evolutionary significance of cis-regulatory mutations. *Nat Rev Genet.* 8(3):206-216,
47 doi:10.1038/nrg2063.
- 48 Wu Y, Popovsky-Sarid S, Tikunov Y, Borovsky Y, Baruch K, Visser RGF, Paran I, Bow A. 2023.
49 CaMYB12-like underlies a major qtl for flavonoid content in pepper (*Capsicum annuum*) fruit. *New*
50 *Phytol.* 237(6):2255-2267, doi:10.1111/nph.18693.
- 51 Wu Y, Wen J, Xia Y, Zhang L, Du H. 2022. Evolution and functional diversification of R2R3 MYB
52 transcription factors in plants. *Hort Res.* 9:uhac058, doi:10.1093/hr/uhac058.
- 53 Xu C, Tachmazidou I, Walter K, Ciampi A, Zeggini E, Greenwood CM, UK10K Consortium. 2014.
54 Estimating genome-wide significance for whole-genome sequencing studies. *Genet Epidemiol.*
55 38: 281-290, doi: 10.1002/gepi.21797.

- 1 Yang Y, Smith SA. 2014. Orthology inference in nonmodel organisms using transcriptomes and low-
2 coverage genomes: Improving accuracy and matrix occupancy for phylogenomics. *Mol Biol Evol.*
3 31(11):3081-3092, doi:10.1093/molbev/msu245.
- 4 Yang J, Wu X, Aucapiña C, Zhang D, Huang J, Hao Z, Zhang Y, Ren Y, Miao Y. 2023. *NtMYB12* requires
5 for competition between flavonol and (pro)anthocyanin biosynthesis in *Narcissus tazetta* tepals.
6 *Mol Hort.* 3: 2, doi:10.1186/s43897-023-00050-7.
- 7 Yarahmadvov T, Robinson S, Hanemian M, Pulver V, Kuhlemeier C. 2020. Identification of transcription
8 factors controlling floral morphology in wild *Petunia* species with contrasting pollination
9 syndromes. *Plant J.* 104(2):289-301, doi:10.1111/tpj.14962.
- 10 Yuan YW, Rebocho AB, Sagawa JM, Stanley LE, Bradshaw HD. 2016. Competition between anthocyanin
11 and flavonol biosynthesis produces spatial pattern variation of floral pigments between mimulus
12 species. *Proc. Natl. Acad. Sci. U.S.A.* 113(9):2448-2453, doi:10.1073/pnas.1515294113.
- 13 Zheng Y, Chen Y, Liu Z, Wu H, Jiao F, Xin H, Zhang L, Yang L. 2021. Important roles of key genes and
14 transcription factors in flower color differences of *Nicotiana glauca*. *Genes (Basel).* 12(12):1976,
15 doi:10.3390/genes12121976.
- 16 Zhou X, Stephens M. 2012. Genome-wide efficient mixed-model analysis for association studies. *Nat*
17 *Genet.* 44(7):821-824, doi:10.1038/ng.2310.
- 18

1 **Table 1.** Phenotypes and genotypes of sampled individuals from backcross population.

Indiv	DEL	CYAN	PEL	<i>F3'H</i> expression	Inferred <i>T</i> -locus genotype	<i>F3'5'h</i>	<i>Dfr</i>
GCG22	0.2	0.68	0.1	high	<i>Tt</i>	<i>F</i> -	<i>dd</i>
GCG55	0.3	0.63	0.1	high	<i>Tt</i>	<i>F</i> -	<i>dd</i>
GCG11	0.24	0.61	0.15	high	<i>Tt</i>	<i>F</i> -	<i>dd</i>
GCG98	0.2	0.7	0.1	high	<i>Tt</i>	–	<i>Dd</i>
GCG84	0.03	0.67	0.30	high	<i>Tt</i>	–	<i>Dd</i>
GCG25	0.2	0.62	0.2	high	<i>Tt</i>	–	<i>Dd</i>
GCG49	0.12	0.53	0.35	high	<i>Tt</i>	<i>F</i> -	<i>Dd</i>
GCG40	0.11	0.67	0.22	high	<i>Tt</i>	<i>F</i> -	<i>Dd</i>
GCG94	0.3	0.57	0.2	(this study)	<i>Tt</i>	<i>F</i> -	<i>Dd</i>
GCG60	0.01	0.70	0.29	high	<i>Tt</i>	–	<i>dd</i>
GCG18	0.01	0.84	0.14	high	<i>Tt</i>	–	<i>dd</i>
GCG76	0.05	0.81	0.14	(this study)	<i>Tt</i>	–	<i>dd</i>
GCG2	0.24	0.07	0.69	low	<i>tt</i>	<i>F</i> -	<i>Dd</i>
GCG61	0.2	0.14	0.7	low	<i>tt</i>	<i>F</i> -	<i>Dd</i>
GCG23	0.21	0.11	0.67	low	<i>tt</i>	<i>F</i> -	<i>Dd</i>
GCG24	0.2	0.14	0.7	low	<i>tt</i>	<i>F</i> -	<i>dd</i>
GCG73	0.2	0.12	0.7	low	<i>tt</i>	<i>F</i> -	<i>dd</i>
GCG7	0.17	0.14	0.69	low	<i>tt</i>	<i>F</i> -	<i>dd</i>
GCG4	0.05	0.05	0.90	low	<i>tt</i>	–	<i>Dd</i>
GCG85	0.02	0.04	0.94	(this study)	<i>tt</i>	–	<i>Dd</i>
GCG6	0.08	0.11	0.81	low	<i>tt</i>	–	<i>Dd</i>
GCG9	0.05	0.04	0.91	low	<i>tt</i>	–	<i>dd</i>
GCG104	0.1	0.08	0.9	low	<i>tt</i>	–	<i>dd</i>
GCG43	0.07	0.06	0.87	low	<i>tt</i>	–	<i>dd</i>

2 The DEL, CYAN and PEL columns show the proportion of anthocyanins derived from blue delphinidin,
3 purple cyanidin and red pelargonidin pigments, respectively (data from Smith and Rausher 2011). The
4 expression of *F3'H* was quantified with qPCR in Smith and Rausher (2011); individuals with 'low'
5 expression have 10-fold lower expression than those with 'high'. Individuals with high *F3'H* expression
6 and primarily cyanidin production are predicted to be heterozygous at the *T*-locus with one 'blue' and one
7 'red' allele (*Tt*). Three individuals were not included in the previous qPCR experiment; their inferred *T*-
8 locus genotype was based on pigment production (CYAN vs. PEL) and their *F3'H* expression was
9 measured as part of this study. Thus, the samples are split between *Tt* and *tt* individuals at the *T*-locus,
10 and there are three replicates for each combination of genotypes at the other involved loci (*F3'5'h* and

Dfr). Note that the red parental species is missing the functional copy of *F3'5'h*, so the red allele is indicated with a -.

Figure Legends

Figure 1. Flavonoid pigment production in parental lines and experimental design for identifying the *T*-locus. a) Segregating backcross populations were created from parental lines of the blue-flowered *Lochroma cyaneum* and the red-flowered *I. gesnerioides*. The former makes delphinidin-derived anthocyanins and all three of flavonols (kaempferol, quercetin and myricetin) while the latter makes only pelargonidin-derived anthocyanins and kaempferol. The active branches of the pathway are shaded in each case. The enzymes shown (in ellipses) are flavonoid 3'-hydroxylase (*F3'H*), flavonoid-3'-hydroxylase (*F3'5'H*), dihydroflavonol reductase (*DFR*) and flavonol synthase (*FLS*). Flavonoid intermediates are dihydrokaempferol (*DHK*), dihydroquercetin (*DHQ*) and dihydromyricetin (*DHM*). Additional steps upstream of *DHK* (e.g. involving chalcone synthase (*CHS*), chalcone isomerase (*CHI*) and flavanone hydroxylase (*F3H*)) and downstream of *DFR* (e.g., involving anthocyanidin synthase (*ANS*), glucosyltransferase) are not shown but indicated with the dashed portion of the arrows. Note that *F3'5'H* has 3' activity and can act on *DHK* in some taxa, but in *Lochroma*, it is specialized for *DHQ* (Smith and Rausher, 2011). The F1 hybrid produces mainly cyanidin-derived anthocyanins and is presumed to be heterozygous at the *T*-locus, which controls *F3'H* expression. b) Phenotypes and pools for RNASeq experiment. We divided the backcross population (F1 crossed to the red parent) into a high *F3'H* expression purple-flowered cyanidin-producing pool (presumably *Tt*) and a low *F3'H* expression mostly or entire pelargonidin-producing pink to red-flowered pool (presumably *tt*). See Table 1 for more information on sequenced individuals.

Figure 2. Manhattan plot showing significant SNP associations on *Lochroma cyaneum* chromosome 5. The blue-dashed line marks the cutoff for genome-wide significance ($P < 5 \times 10^{-8}$). The gold vertical line marks the location of the *MYB12-like* gene, which predicts *F3'H* expression. a) The position of *MYB12-like* near the 3' prime end of the chromosome. b) Close-

up of the region containing *MYB12-like*, showing the concentration of associated SNPS in the last 500kb of the chromosome.

Figure 3. Co-expression of *F3'H* and *lcMYB12-like*. a) Expression levels for each gene in the two phenotypic pools. Box plots mark the first and third quantiles, with a bisecting line to indicate the median. The mean is denoted with an x. The Pearson correlation coefficient for the expression of these two genes is 0.91187 ($P = 1.58 \times 10^{-05}$ after Bonferroni correction). b) Histogram of Pearson correlation coefficients between *F3'H* and all other expressed genes. The orange vertical bar marks the value for the correlation between *F3'H* and *MYB12-like*. The value for the correlation with *FLS* also falls at that mark ($r = 0.91185$). See Supplementary Table 4 for complete ranked list of correlation coefficients. c) Submodule from WGCNA analysis containing all edges including *F3'H* and *MYB12-like* (Supplementary Table 5). The lines representing each edge are colored by the connectivity value (TOM) from WGCNA (Fig. S6); more closely clustered genes are more tightly co-expressed (i.e., spring layout). Genes encoding enzymes related to the flavonoid pathway (CHS, FLS, UGT) are outlined along with *lcMYB12-like*.

Figure 4. Phylogenetic relationships among Solanaceae subgroup 7 (SG7) MYBs. a) Maximum likelihood phylogeny from amino acid sequences; the sequences highlighted in light yellow include the best studied SG7 MYBs in Solanaceae, e.g., *S/MYB12* (Tomato.01g079620) that is tied to the colorless fruit epidermis phenotype (Ballester et al. 2010). The deeply diverged *MYB12-like* sequences are highlighted in dark yellow; *lcMYB12-like* corresponds to *lochroma.IC05g034110*. Bolded branches have >95% bootstrap support; values between 50 and 95% bootstrap support are given. The outgroups used to root the tree (SG7 MYBs from *Arabidopsis*) have been pruned. b) Maximum likelihood phylogeny for Solanaceae *MYB12-like* sequences based on a complete CDS alignment. Bootstrap supports are shown as in (a). Tip values for *MYB12-like* expression (TPM) and floral flavonol content (in mg/g from Larter et al.

1 2019) for six species are colored by magnitude (see Supplementary Table 7 for raw data).
2 These data are graphed in the inset figure with the dashed line showing the linear trend.
3 *lochroma cyaneum* is not included as data from previous transcriptomic analyses (Gates *et al.*
4 2018) are not directly comparable with the *de novo* transcriptomes from the present study. Full
5 names and sources for all sequences used in these analyses are given in Supplementary Table
6 6. Branch lengths in both trees are in substitutions per site.

7

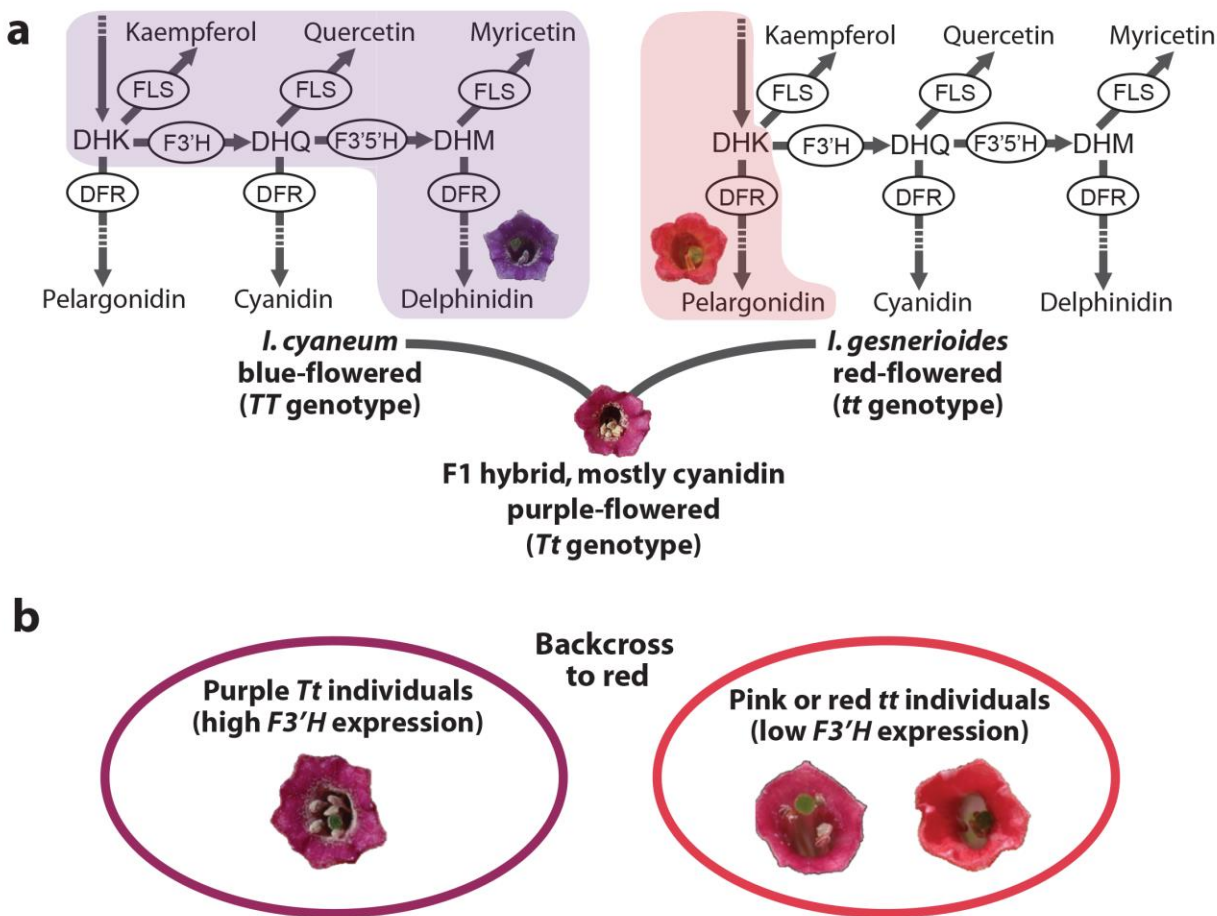


Figure 1
165x125 mm (x DPI)

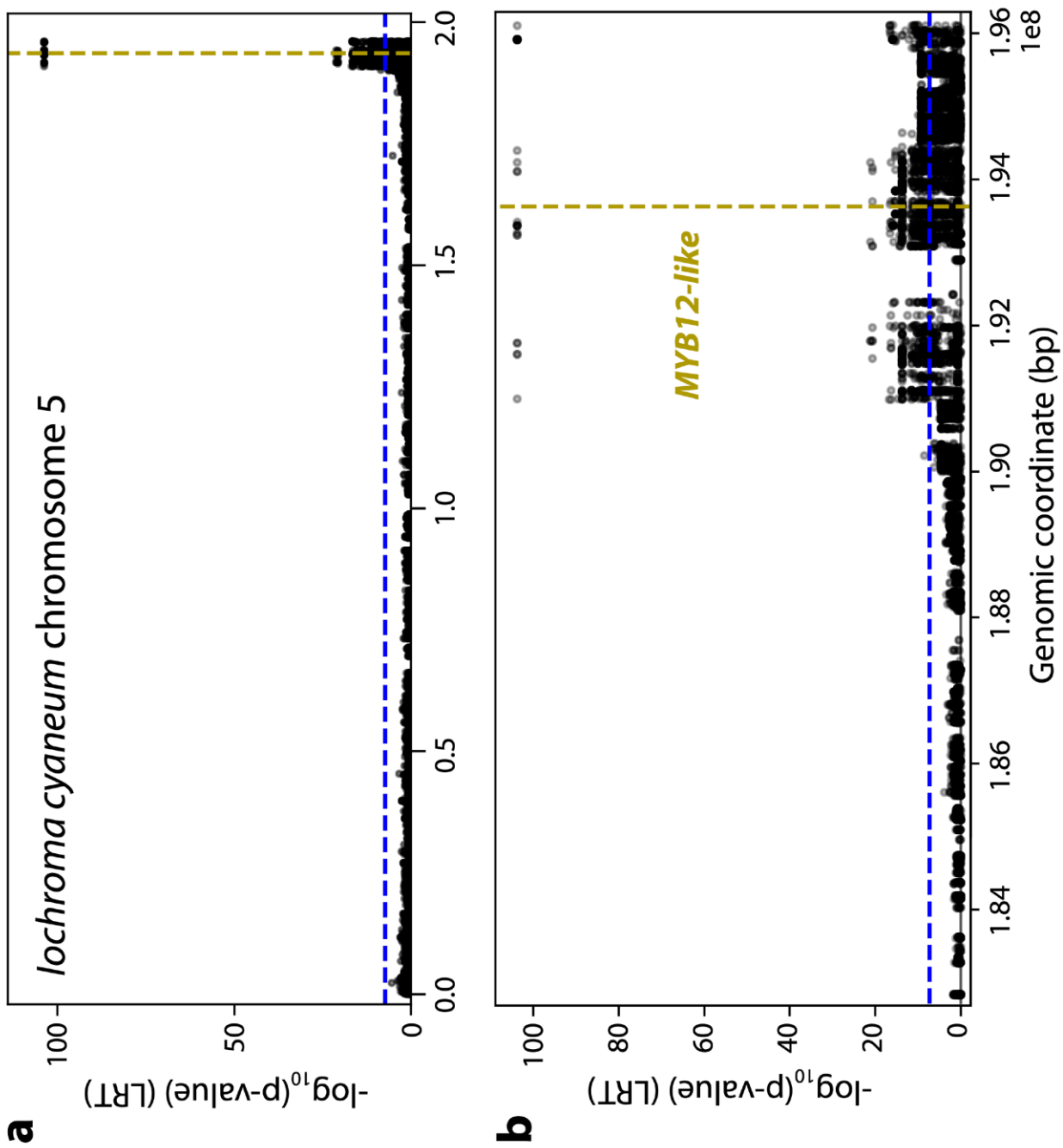


Figure 2
165x178 mm (x DPI)

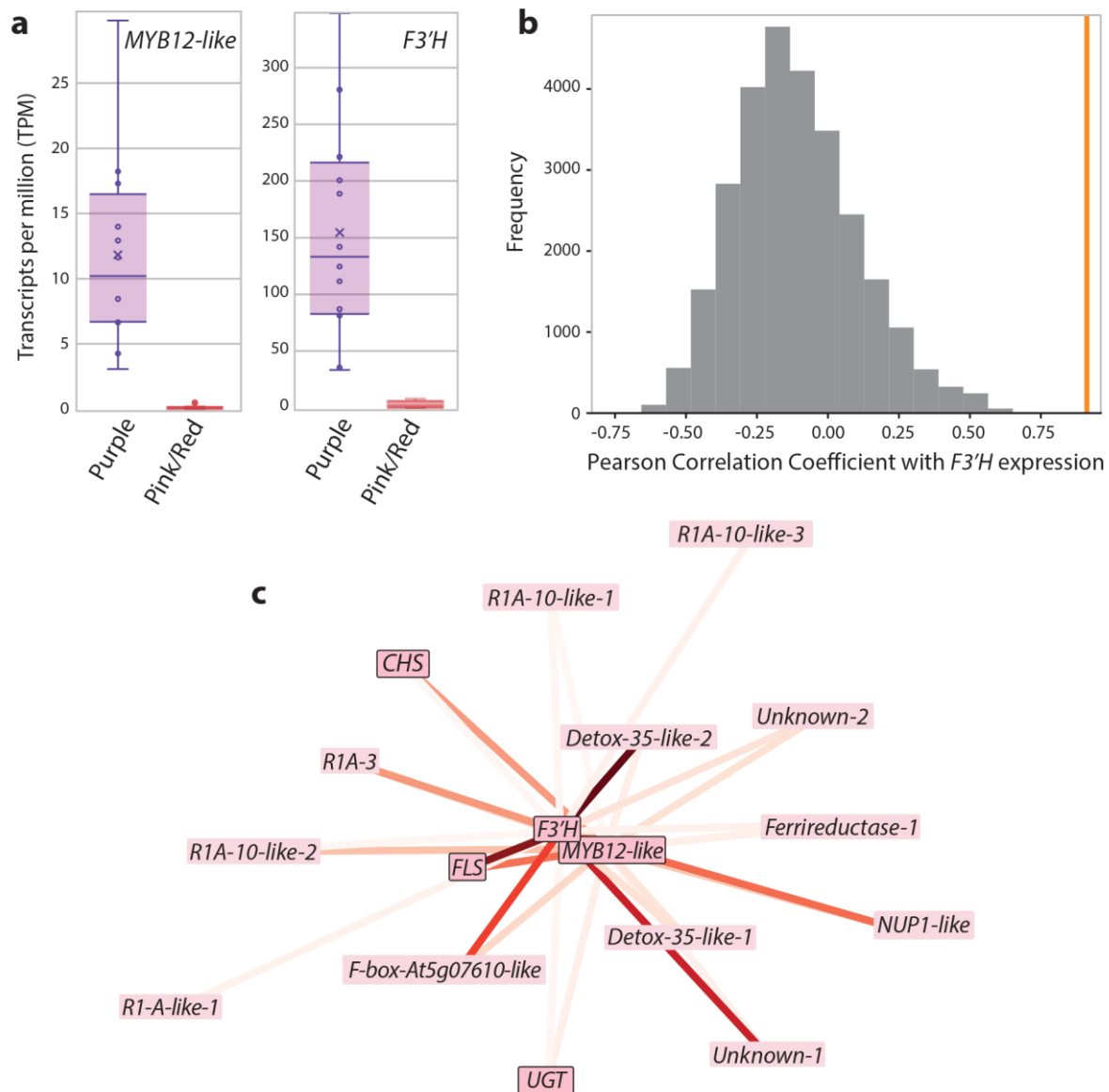


Figure 3
158x156 mm (x DPI)

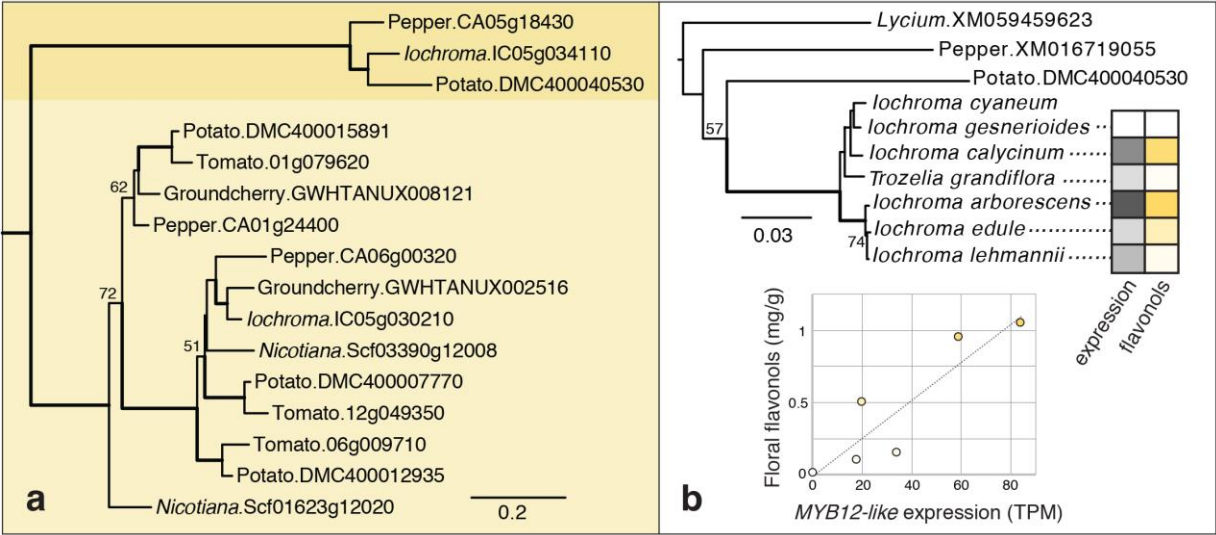


Figure 4
165x72 mm (x DPI)